

Drift-Sense — Phase 2 Failure Analysis

Registration under Unknown Pose — addendum to the Phase 1 submission. This document covers what is NEW for Phase 2: four failure modes found and fixed during development, the validation evidence for each fix, and the limitations that remain and are disclosed rather than hidden.

1. Rotation sign convention (verified, not a bug)

Before trusting the existing pipeline's rotation output for a newly-scored quantity, we checked it directly: generated pairs with a known capture rotation, swept candidate template angles, and took the argmax. The angle that maximizes correlation is consistently the NEGATIVE of the generator's own rotation parameter, and matches the addendum's stated convention (CCW-positive, reference as it appears in the search) once combined with OpenCV's own documented `+angle=CCW` rule (independently confirmed with a synthetic test image). Net finding: the existing angle-search machinery already reports the correct value — no change needed to the search itself, only to how our own synthetic ground truth is compared against it.

2. Scale quantization (real bug, fixed)

The existing template builder resizes to $\text{round}(\text{reference_size} / \text{scale})$ — an INTEGER pixel count. Sweeping scale at a known true value showed a visible staircase: many adjacent scale values collapse onto the same integer template size and produce identical correlation, with plateau width $\sim 0.06\text{--}0.15$ at this project's 8–12x range. That is the same order as Phase 2's 1% top-credit tolerance ($\sim 0.08\text{--}0.12$ absolute) — no optimizer on top of this can resolve finer than the plateau it's standing on. Fix: a second template builder samples a FIXED-size canvas via one continuous affine transform (with scale-matched Gaussian pre-filtering to avoid aliasing), so scale sensitivity is genuinely continuous. Used only in the new local pose-refinement stage; the coarse/fine grid stages are untouched.

3. Local refinement jumping to a competing periodic repeat (real bug, fixed)

The new local pose-refinement step crops a window around an already-chosen candidate site to polish scale/rotation/subpixel position. That crop is sized off the TEMPLATE, which can be much larger than the pattern's own period at 8–12x scale (e.g. template $\sim 100\text{px}$ vs. period $\sim 7\text{--}13\text{px}$) — so the crop can contain a dozen-plus repeats, and an unconstrained argmax inside it can silently jump to a neighboring repeat that scores marginally higher from noise, undoing the periodic-ambiguity tie-break that had already run. Diagnosed precisely: a failing case's output was off from the intended site by EXACTLY 4 and 8 whole periods in x and y. Fix: the refinement's internal search is now windowed to a few pixels around the intended anchor. On the exact 20-trial pure-periodic test that first exposed this, accuracy against the correct (nearest-to-centre, see §5) target went from 0% to 85% within 5px (65% within 1px), with no regression on the landmark-bearing precision trials that validated fix #2 (100% within the tightest scale/rotation credit tier, 7/7).

4. Rejection F1's positive class (verified against the addendum's own claim)

The addendum states a team that never rejects anything scores zero on the 15-pt rejection metric. Computing F1 with "found" as the positive class gives ~ 0.875 for that strategy, not zero. Computing it with "reject" (found=0, i.e. correctly detecting absence) as positive gives exactly 0.0 — matching the addendum exactly. This determines which class the metric is defined over, which in turn changes where the found-threshold should sit; our threshold was re-tuned against this corrected definition, and additionally validated against a full simulation of the 100-pt formula (not F1 in isolation), since a false negative on a present pair also forfeits that pair's entire localization+pose credit under the output contract (found=0 writes 0 in the pose columns).

5. Known, disclosed limitations

- Landmark-free ("pure periodic") present pairs: the true site is inherently ambiguous among many equally-valid periodic repeats. We score these against the nearest-repeat-to-centre target implied by the addendum's own tie-break rule, not the raw random placement our synthetic generator draws for testing.
- dram_arcuate's tilted rhombic lattice is only approximately captured by an axis-aligned period model (inherited from the Phase 1 codebase's own documented caveat) — its nearest-to-centre comparisons run a few pixels less precise than dram/finfet's in our validation.
- finfet absent pairs are the hardest rejection case: two independent finfet realizations (different period/linewidth draws) can still correlate fairly strongly given the style's simple, repetitive geometry, more so than dram or dram_arcuate. This is the main remaining drag on rejection F1.
- The addendum's own pairs.csv format is not yet published (arrives T+2); register.py detects common column-name variants and falls back to this project's own references/<id>.png convention, but this should be verified against the real 3 sample pairs the moment they arrive, before the T+7 freeze.

6. Validation snapshot

Metric	Result	Notes
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Scale error, landmark-bearing (7 controlled trials)	100% within 1% (top credit tier)	post-fix #2
Rotation error, landmark-bearing (7 controlled trials)	100% within 0.25° (top credit tier)	post-fix #2
Pure-periodic localization vs. nearest-to-centre target (20 trials)	85% within 5px, 65% within 1px	post-fix #3, was 0%
Set A / Set B loc. credit (held-out sim, two independent seeds)	~0.85–1.00 / ~0.70–0.73	evaluate_p2.py
Set D (optical/RGB bonus) loc. credit	1.00 on all spot-checked pairs	evaluate_p2.py
Rejection F1 (reject-positive, corrected)	0.48–0.75 across seeds	weakest component; \$5
Median / max wall clock per pair	~1.0–1.3s / <2s	budget: median≤5s, hard cap 20s

Method unchanged from Phase 1's declared approach: classical multi-scale/rotation NCC with a coarse-to-fine pyramid, a small from-scratch NumPy CNN for landmark re-ranking, and FFT-autocorrelation lattice enumeration for periodic tie-breaks — extended with continuous pose refinement and a calibrated presence/score classifier, both additive. All four items above were found by direct measurement against known ground truth, not assumed; docs/validation_report_v4.md has the full derivations and code references.